Abijeet Rajasekaran

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Education

Vellore Institute of Technology Sept 2022 to Aug 2026
Bachelor of Technology in Computer Science and Engineering | CGPA: 8.93/10 *Vellore, India*

- **Chancellor's Achiever Award:** Awarded for outstanding merit and national-level achievements representing the institute across research, debate, and social work.

Research Experience

Invited Machine Learning Research Fellow *(PI: Prof. Somdatta Goswami)* Aug 2026 to Present
Johns Hopkins University, Centrum IntelliPhysics *Baltimore, MD*

- Studying uncertainty in learned dynamical models used within sequential inference, embedding independently trained TI-DeepONet time-stepping models in ensemble Kalman filtering for chaotic dynamical systems such as Kuramoto–Sivashinsky.
- Investigating whether disagreement among independently trained models reflects their actual forecast error, and how to separate shared model bias and per-step learned-model error from amplification by the underlying chaotic dynamics.
- Accelerating the group's differentiable JAX finite-element PDE solver through custom sparse CUDA kernels and multigrid preconditioners, with the goal of reducing the computational cost of ensemble and Bayesian scientific-ML methods.

Machine Learning Research Intern *(PI: Prof. Jinwoo An)* Dec 2025 to June 2026
University of Texas Rio Grande Valley *Edinburg, TX*

Calibrated Surrogate Modeling and Uncertainty Quantification

- Built ensemble surrogate models for sparse Engineered Cementitious Composite data and replaced point estimates with distribution-free conformal prediction intervals carrying finite-sample coverage guarantees.
- Decomposed predictive uncertainty into epistemic and aleatoric components, with approximately 68% attributed to epistemic uncertainty, indicating that missing information rather than irreducible measurement noise dominated predictive uncertainty.
- Incorporated physical strain-hardening constraints and evaluated sensitivity to input, process, and cost variability using Monte Carlo simulation.

Vision-to-Physics-Informed Learning for Fracture Mechanics

- Independently proposed and built a computer-vision-to-PINN pipeline for estimating stress-intensity factors from crack geometry and diagnosed spectral-bias failure near the crack-tip singularity.
- Replaced the standard residual formulation with a variational potential-energy objective and J-integral readout, achieving 0.72% maximum relative error against validated finite-element references across approximately 20 runs while halving run-to-run variation.

Undergraduate Research Lead *(PI: Prof. Yuvaraj Natarajan, Prof. Bubryur Kim)* May 2025 to Dec 2025
VIT and Kyungpook National University, CRAFT Autonomous Hazard-Mapping Rover *Vellore / Remote*

- Served as the first project member and research lead, recruiting and mentoring a six-member team developing perception and mapping systems for autonomous inspection of fire, smoke, chemical spills, and other hazards.
- Developed and deployed a six-class YOLOv8-Nano perception system reaching mAP@0.5 of 0.845 and 24.3 ms TensorRT FP16 inference on a Jetson Orin Nano.
- Fused heterogeneous sensor streams using a Kalman filter whose noise levels had to be manually specified because the learned perception system provided no calibrated uncertainty; this problem motivated my subsequent work in uncertainty quantification.

Research Outputs

- B. Kim, **A. Rajasekaran**, A. Kaleeshwar, S. Preethaa, Y. Natarajan, J. An. *Uncertainty-Aware Machine Learning and Inverse Mix Design for Engineered Cementitious Composites.* Manuscript under submission.
- **A. Rajasekaran**, A. Kaleeshwar, J. An, Y. Natarajan. *Vision-Guided Williams-Enriched Physics Informed Neural Network for Mesh-Free Stress Intensity Factor Assessment of Cracked Concrete.* Manuscript under submission.
- B. Kim, **A. Rajasekaran**, A. Kaleeshwar, S. Preethaa, Y. Natarajan. *Cross-modal adaptive exploration and chemo-visual hazard-aware navigation for autonomous robotic inspection in hazardous environments.* Manuscript under submission.
- C. Verma, A. Kaleeshwar, **A. Rajasekaran**, G. Dhanasekaran, Y. Natarajan. *Autonomous Robotic System for Gaseous Hazard Mapping and Structural Damage Detection.* Indian Patent Application No. 202641072251. Published.

Teaching, Mentoring and Outreach

Board Member Jan 2024 to Jan 2025
Debate Society, Vellore Institute of Technology *Vellore, India*
- Led the society's editorial domain and trained junior cohorts in argumentation, adjudication, reasoning from first principles, and technical writing; competed nationally and internationally in varsity debate.

Executive Member and Documentation Head July 2024 to Dec 2024
A Responsible Citizen Foundation, Educational Outreach NGO *Remote*
- Taught mathematics and computer science to high-school students and raised funds for educational outreach in underprivileged areas in the Indian west zone.

Educational Mentor June 2024 to Feb 2025
Department of Student Welfare, Vellore Institute of Technology *Vellore, India*
- Conducted STEM drives for students at government high schools in Vellore.

Industry Experience

Generative AI Intern June 2025 to Aug 2025
CyrusLogic Pvt. Ltd. *Bangalore, India*
- Built AWS-based LLM pipelines using Lambda and S3, local inference with Ollama, and n8n agentic workflows; reduced extraction latency by 20%.

Data Science Intern May 2025 to June 2025
RippleThrive Pvt. Ltd. *Hosur, India*
- Deployed Transformer-based text-analysis systems and built automated data pipelines integrating scraped content with Slack and social-media APIs.

Software Development Intern May 2024 to July 2024
Karomi Pvt. Ltd. *Chennai, India*
- Built a Three.js simulator that transforms flat packaging dielines into articulated 3D models using vector geometry and rigid-body transformations.

Technical Skills

Probabilistic ML and UQ: Ensemble Kalman filtering, conformal prediction, deep ensembles, Monte Carlo methods, Bayesian inference, uncertainty decomposition
Scientific ML: Neural operators, DeepONet, physics-informed neural networks, data assimilation, finite-element methods, spectral methods, Krylov solvers (conjugate gradient) with algebraic multigrid (AMG) and Jacobi preconditioning
Scientific Computing: PyTorch, JAX, FEniCS, CUDA, multigrid methods, TensorRT, OpenCV
LLM and Agentic Systems: Local LLM inference with Ollama, RAG, n8n agent workflows, AWS Lambda/S3, Transformers
Programming: Python, C/C++, R, SQL, JavaScript, LaTeX